

HAT-P-36b

R-0880

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240228 · created 2026-07-29T00:38:06+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460368.8426 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.046
Transit depth (Rp/Rs)^2	2.45 +/- 1.43 [%]
Orbital Inclination (inc)	81.39 +/- 2.86
Airmass coefficient 1 (a1)	0.46 +/- 0.29
Airmass coefficient 2 (a2)	0.64 +/- 1.36
Scatter in the residuals of the lightcurve fit is	59.04 %
Best Comparison Star	None
Optimal Aperture	1.95
Optimal Annulus	7.76
Transit Duration (day)	0.079 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.046 vs 0.126 (deviation 0.67σ)
Duration fit vs published	0.079 d vs 0.0925 d (deviation 0.71σ)
Mid-time O–C	-10.8 min
Depth SNR	3.4
Residual scatter	59.04 %

Light curve

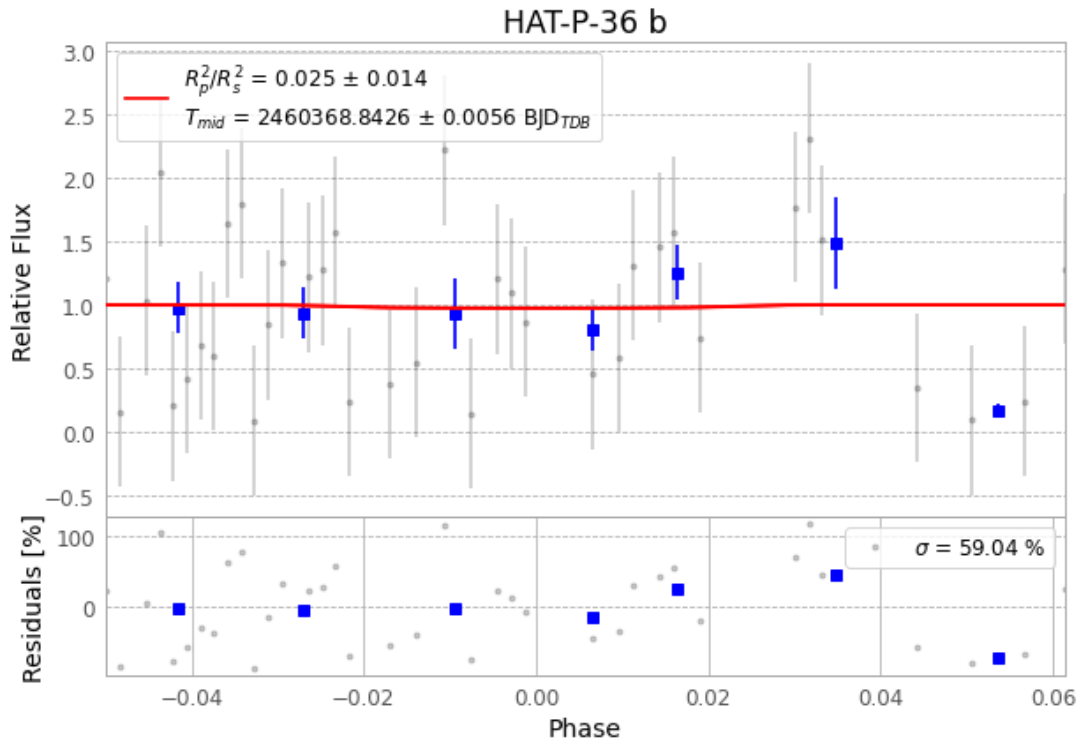


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-02-27.png (SHA-256 1aba4cae24fb42c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 260], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5d2c1e7654bdc0c4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

80 FITS frames (53 MB), HATP-36240228062715.FITS → HATP-36240228102415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240228.log (d62926cb1d0a...), FinalLightCurve_HAT-P-36 b_2024-02-27.png (1aba4cae24fb...), FinalLightCurve_HAT-P-36 b_2024-02-27.csv (a03ef639afc2...)

Compute resources

Wall time 245.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 00:38Z.